



**Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 8 • STANDARD 6.AT.9

# Write and Represent Inequalities

Lesson 8-4 • Teacher Copy

**Name:** \_\_\_\_\_

**Date:** \_\_\_\_\_

**Class:** \_\_\_\_\_

## My Notes

Level 2 Standard



### TODAY'S OBJECTIVES

**Content Objective:** I can write inequalities to represent real-world situations.

**Language Objective:** I can explain my inequality using the words inequality, greater than, less than, and at least / at most.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



### WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Inequality

Greater than

Less than

At least / At most

Variable

No more than

Constraint

Solution set



Tap any word to see its meaning and a picture.

1

A math sentence comparing values with  $<$ ,  $>$ ,  $\leq$ , or  $\geq$  is an

2

The symbol  $>$  that means one value is larger than another means

3

The symbol  $<$  that means one value is smaller than another means

4

'At least' means greater than or equal to, and 'at most' means

 than or equal to.

5

A letter that stands for an unknown number is a

6 The phrase that means less than or equal to is "no more than," written with the  symbol.

7 A limit that every solution must satisfy is a .

8 All the values that make an inequality true form its .

## Write About the Math

**What speeds are possible for the getaway car? How did the phrase 'at most' help you write the inequality?**

*¿Qué velocidades son posibles para el carro de huida? ¿Cómo te ayudó la frase «como máximo» a escribir la desigualdad?*

Say your answer to a partner first. Then write it, using at least two of these words:

☐ **Inequality**  
*Desigualdad*

☐ **Greater than**  
*Mayor que*

☐ **Less than**  
*Menor que*

☐ **At least / At most**  
*Al menos / A lo más*

☐ **Variable**  
*Variable*

**Model:** The age could be 18 or any number greater, so it is not a single value. — Students explain that 'at least 18' allows many values (18 and up), so an inequality with  $\geq$  captures it while an equation would force one value.

### Start

Complete the frames. Use the word bank.

*Completa las oraciones. Usa el banco de palabras.*

- A speed of 65 mph is \_\_\_\_, a speed of 70 mph is \_\_\_\_, because 'at most 65' means the speed must be \_\_\_\_ 65.
  - My answer is \_\_\_\_ because \_\_\_\_.
- Mi respuesta es \_\_\_\_ porque \_\_\_\_.*
- 
- 

### Build

Write two connected sentences. Use because, so, or but.

*Escribe dos oraciones conectadas. Usa porque, entonces o pero.*

- My claim is \_\_\_\_.
- Mi afirmación es \_\_\_\_.*
- My evidence is \_\_\_\_, so \_\_\_\_.
- Mi evidencia es \_\_\_\_, entonces \_\_\_\_.*

---

---

---

### Explain

Write a claim, give math evidence, and explain your reasoning.

*Escribe una afirmación, da evidencia matemática y explica tu razonamiento.*

---

---

---

---

---

☐ I answered the question.

☐ I used math evidence: numbers, an equation, or a model.

☐ I used at least two math words.

---

## Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

---

## Answer Key & Teacher Guide

1. **Notes 1:** Inequality
2. **Notes 2:** Greater than
3. **Notes 3:** Less than
4. **Notes 4:** At least / At most
5. **Notes 5:** Variable
6. **Notes 6:** No more than
7. **Notes 7:** Constraint
8. **Notes 8:** Solution set
9. **Watch for this mistake:** *A common mistake in Write Inequalities is using a strict symbol ( $>$  or  $<$ ) for a phrase that actually includes the boundary value. For 'the suspect is at least 18 years old,' students often write  $a > 18$  instead of the correct  $a \geq 18$  — this wrongly excludes 18 itself, even though 'at least 18' means 18 counts as a match. The same error happens in reverse with 'at most': for 'the getaway bag holds at most 12 items,' students write  $n < 12$  instead of  $n \leq 12$ . Always ask: does the key phrase include the number ('at least,' 'at most,' 'no more than,' 'no fewer than'  $\rightarrow$  use  $\geq$  or  $\leq$ ) or exclude it ('more than,' 'fewer than'  $\rightarrow$  use  $>$  or  $<$ )?*
10. **Try It 1:** A.  $g \geq 12$  — 'At least 12' includes 12 and any number above it, so  $g \geq 12$ .
11. **Try It 2:** A.  $n < 5$  — 'Fewer than 5' does not include 5, so  $n < 5$ .